

Karanveer Singh

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Education

Simon Fraser University, Burnaby, BC, Canada

Expected graduation: June 2028

Bachelor of Science in Computing Science

Relevant Courses: Data Structures & Algorithms, Software Engineering, AI, Database Systems

Technical Skills

Languages: Java, TypeScript, JavaScript, Python, SQL, C++

Web/Backend: React, Next.js, REST APIs, MongoDB, HTML, CSS

Testing/Tools: JUnit, JaCoCo, Git, GitHub

AI/ML: scikit-learn, pandas, NumPy

Experience

Freelance Web Developer

Sep 2025 – Present

Self-Employed

- Designed and delivered responsive static websites for local trade businesses, improving online presence through clear service pages, gallery sections, and contact flows
- Worked directly with business owners to gather requirements, organize service content, revise layouts, and prepare sites for public use
- Built mobile-friendly service pages, calls-to-action, contact sections, and visual layouts using HTML and CSS

Robotics Mentor

Sep 2022 – April 2023

FIRST

- Mentored a team of 6 students in robotics programming and mechanical design using LEGO Mindstorms, improving their ability to design autonomous robots and write structured control logic.
- Led the team to 2nd place in the regional FIRST LEGO League finals, improving robot performance and competition strategy.

Projects

MarketPulse | [GitHub](#)

May 2026

Next.js, TypeScript, React, MongoDB, Finnhub API, Inngest, Gemini

- Built a full-stack stock market dashboard with authenticated user accounts, live stock search, stock detail pages, and MongoDB-backed watchlists
- Implemented price alert workflows using Finnhub quotes, MongoDB alert storage, Inngest background jobs, and email-style notification logic
- Integrated Gemini AI to generate personalized welcome messages and daily market summary briefings based on each user's watchlist

Haunted Hotel | [GitHub](#)

September 2025 – December 2025

Java, Maven, JUnit, JaCoCo

- Developed gameplay systems for a 2D tile-based game including enemy AI, combat, and event-driven progression
- Implemented A* pathfinding and collision systems to improve game responsiveness and behavior
- Wrote unit and integration tests using JUnit with coverage tracked via JaCoCo

Spam Detection System | [GitHub](#)

January 2026 – April 2026

Python, scikit-learn, pandas

- Achieved 97.7% accuracy with zero false positives on a held-out test set of 1,115 messages using an 80/20 train-test split
- Built a spam classification pipeline using Logistic Regression with Bag-of-Words and TF-IDF feature extraction
- Evaluated model performance using precision, recall, F1-score, and confusion matrix analysis

Myopia Progression Prediction Model | [GitHub](#)

December 2025 – March 2026

Python, scikit-learn, Streamlit

- Built an end-to-end ML pipeline to predict long-term myopia progression using lifestyle and patient data
- Engineered domain-specific features to capture non-linear effects and improve prediction quality
- Achieved 0.40 MAE on held-out test data through optimized feature design and model tuning

AutoCurve Assistant | [GitHub](#)

January 2026

Python, FastAPI, JavaScript, scikit-learn, OpenRouter

- Built an AI-powered car valuation app that estimates used-car prices using regression models trained on 52K+ vehicle listings
- Integrated image-based vehicle condition scoring with structured car data to generate adjusted price estimates
- Evaluated model performance on held-out test data, achieving $R^2 = 0.84$ for price prediction